

Micro-Renovator

Bringing Processor Firmware up to Code



Bio

Matt (a.k.a. Syncsrc)

- Recovering EE / CompE
- Builder and user of hardware debug features
- Uses “BIOS” and “UEFI” interchangeably
- Currently responsible for platform security of a cloud
- Religiously updates firmware
- Formerly a product security validation lead at Intel

Background



Meltdown

Meltdown breaks the most fundamental isolation between user applications and the operating system. This attack allows a program to access the memory, and thus also the secrets, of other programs and



Spectre

Spectre breaks the isolation between different applications. It allows an attacker to trick error-free programs, which follow best practices, into leaking their secrets. In fact, the attack bypasses all the checks of said best practices.

the **INQUIRER**

Artificial Intelligence Internet of Things Open Source Hardware Software Security

etmaster hack > iOS 12 > Galaxy Note 9 > iPhone 9 > World Cup >

Intel, ARM and AMD all affected by security-bypassing, kernel-bothering CPU bugs

Fixes exist but it looks like fundamental processor designs are borked

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TECH —

“Meltdown” and “Spectre”: Every modern processor has unfixable security flaws

Immediate concern is for Intel chips, but everyone is at risk.

PETER BRIGHT - 1/3/2018, 4:30 PM

Background - Patches for Spectre & Meltdown

- Kernel Only
 - Meltdown: PTI (Linux), KVA Shadow (Windows)
- Kernel and Application
 - Spectre v1: Lfence insertion
 - Spectre v2: Retpoline (an incomplete fix, per Intel)
- Kernel and Microcode
 - Spectre v2: New SPEC_CTRL and PRED_CMD MSRs
 - Spectre v4: SSBD bit added to SPEC_CTRL MSR
 - *Microcode updates are necessary to expose these new registers*

What is Microcode?

- Can be thought of as “Processor Firmware”
 - A Brief History of Microprogramming
- Can be patched to fix bugs & errata
 - However, processors don't have any non-volatile storage
 - Any applied patches are lost on reset or power-down
- Stored and applied by the *BIOS* or *Operating System*
 - Reloaded on every boot, reset, S3 resume, etc...
 - Intel 64 and IA-32 Architectures SDM Volume 3A, section 9.11.6

Obtaining Microcode Patches

- From BIOS Updates
 - MacBook and Surface users are covered
- From Operating System Updates
 - Linux: redistributed by most distros (via `microcode_ctl.rpm` or `intel-microcode.deb`)
 - Users can also download and install [microcode.dat directly from Intel](#)
 - Microsoft didn't start distributing microcode until 2 months after it was released
 - [Still only distributing for some processor and Windows versions](#)
- *Patches exist, but are impossible to apply to many systems*



FOX

Systems Unable to Patch Spectre

- Still no way to mitigate Spectre v2 on millions of systems
 - Other than buying a new OS (or a new computer)
- Windows PCs that are 3 to 9 years old *
 - Mostly shipped with Windows 7 or 8 pre-installed
 - BIOS updates delayed or unavailable
- 3rd-party microcode update drivers are ineffective

* CPUs more than 9 years old are not receiving microcode updates from Intel

Systems Unable to Patch Spectre

CPU	BIOS Updates	Windows 7 & 8	Windows 10
8th Gen	Available	N/A	None
7th Gen	Available	N/A	None
6th Gen	Available	Negligible	None
5th Gen	Available *	49 Million	None
4th Gen	Available *	114 Million	None
3rd Gen	Xeon Only	141 Million	None
2nd Gen	Xeon Only	149 Million	None
1st Gen	Xeon Only	216 Million	216 Million

Are there any other options?

- When can microcode patches be applied?
 - BIOS - users can't modify
 - OS - microcode drivers run too late
 - Bootloader - *maybe?*
- No existing EFI utility to load microcode
 - TianoCore is open source though
 - And already has code that applies microcode updates
 - How hard could it be?

Uload.efi

- Built using EDK2
- Mostly code appropriated from MicrocodeUpdateDxe
 - Made into an EFI shell app
- Loads microcode to all Processors/Threads

Inserting Uload into EFI Boot

- POST: Power On Stuff That-happens
- Determine bootloader from NVRAM variables
- Locate boot device & partition
 - Involves UUIDs somehow
- Run Bootloader (eg: bootx64.efi, shim.efi)
 - Bootloader is just an EFI application, it can be replaced
- Bootloader launches kernel

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Spectre Status
Speculation control settings for CUE-2017-5715 [branch target injection]
For more information about the output below, please refer to https://support.microsoft.com/en-in/help/4074629

Hardware support for branch target injection mitigation is present: False
Windows OS support for branch target injection mitigation is present: True
Windows OS support for branch target injection mitigation is enabled: False
Windows OS support for branch target injection mitigation is disabled by system policy: False
Windows OS support for branch target injection mitigation is disabled by absence of hardware support: True

Speculation control settings for CUE-2017-5754 [rogue data cache load]

Hardware requires kernel UA shadowing: True
Windows OS support for kernel UA shadow is present: True
Windows OS support for kernel UA shadow is enabled: True
Windows OS support for PCID performance optimization is enabled: True [not required for security]

Suggested actions

* Install BIOS/firmware update provided by your device OEM that
n mitigation.

BTIHardwarePresent      : False
BTIWindowsSupportPresent : True
BTIWindowsSupportEnabled : False
BTIDisabledBySystemPolicy : False
BTIDisabledByNoHardwareSupport : True
KUShadowRequired        : True
KUShadowWindowsSupportPresent : True
KUShadowWindowsSupportEnabled : True
KUShadowPcidEnabled      : True

PS C:\SpeculationControl>
```

InSpectre

Release #8

Check Windows operating system
and processor hardware safety.
Freeware by Steve Gibson / @Sggc

Spectre & Meltdown Vulnerability Status

System is Meltdown protected: **YES**
System is Spectre protected: **NO!**
Microcode Update Available: **YES**
Performance: **GOOD**
CPUID: **406E3**
(full details below)

See GRC's InSpectre webpage at <https://grc.com/inspectre.htm>
for a full explanation of the use and operation of this freeware utility.

Disable Meltdown Protection

Enable Spectre Protection

Exit



Micro-Renovator

- Script to automatically update EFI boot partition
 - Runs from a Linux Live CD
- Finds EFI partition and bootloader
 - Copies microcode and Uload.efi to the boot partition
 - Installs Shell.efi and sets as the primary boot option
 - On boot, startup script runs Uload prior to the OS bootloader

<https://github.com/syncsrc/MicroRenovator>



Limitations

- Breaks Sleep (S3)
 - Hibernation still works
- No secure boot support (yet)
- Occasional inconsistent behavior after booting into Windows
- Microsoft appears to be actively reverting the changes made by earlier versions of MicroRenovator

Summary

- Firmware patching is an unsolved problem
 - UEFI should have made things better, but didn't
- Component vendors needs to focus on enabling patching for end-users
 - Not system builders (they aren't incentivized)
 - IoT and Mobile spaces have the same issue
- *It shouldn't take bootloader hacks to apply security patches to operating systems that are still under support*

Questions?

